

Two-Phase Interleaved DC Charging Method of Integrated Charger for Efficiency Enhancement

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Abstract—Lately, the battery voltage of electric vehicles (EVs) has risen, a twofold increase from the previously commercialized models. However, the adoption of high-voltage batteries has introduced a challenge for the use of lower-voltage fast dc chargers. To address this issue, integrated step-up charging topology that utilizes the motor drive system has been studied. In this paper, a two-phase interleaved integrated charging strategy is proposed. Only two phases of the motor are utilized to boost charge the battery while regulating the generated dc torque to zero. To ensure a standstill condition during charging, the torque produced by the zero-sequence components is also analyzed. The proposed strategy further reduces core losses by diminishing the current ripple in the windings, thereby improving energy efficiency. The feasibility of this work is theoretically assessed, and the experimental results demonstrate the effectiveness of the proposed strategy.

Keywords—dc charging, efficiency, electric vehicle, high voltage battery, integrated charger, interleaved switching, rotation suppression, synchronous motor

I. INTRODUCTION

Recently, the voltages of EV batteries have been doubled to 800 V, which has extended their driving range [1]. However, this elevated battery voltage exceeds the voltage limit of fast dc chargers such as 135 kW Tesla Supercharger, which has a maximum output voltage of 410 V [2]. Additionally, the output voltages of 1.0-1.2 CHAdeMO and FC50 CCS are limited to 500 V [3]. When the voltage limit of a charger is lower than the voltage of a battery, the direct charging method becomes unavailable. To address this situation, step-up dc-dc conversion is required to charge high-voltage batteries with lower-voltage chargers.

Conventionally, an on-board step-up converter is added to utilize widespread lower-voltage fast dc chargers [4]. However, attaching an on-board converter demands extra devices, which increases weight and volume. Consequently, adding a converter becomes an expensive and a range-reducing solution. Instead of adding additional converters, the integrated charger has been introduced, as shown in Fig. 1 [5]–[7]. In integrated charger, the traction system is utilized to charge the battery, with the charger ends connected to the neutral point of the motor and the lower side of the battery. As demonstrated, the integrated charger is structured as a three-phase interleaved boost converter.

A crucial aspect of integrated charger is to keep the rotor at

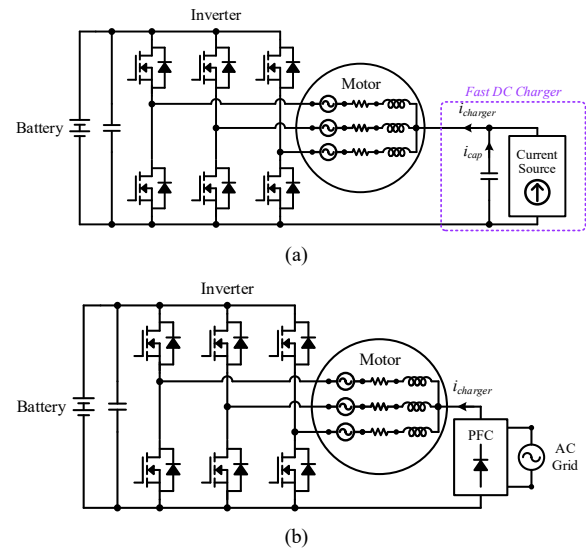


Fig. 1. Schematic of charging systems utilizing the motor windings and a voltage-source inverter. (a) DC charger configuration (b) PFC boost charger configuration

a standstill since the motor windings are utilized for charging. In other words, the stator current must not produce dc torque; otherwise, nonzero torque can cause stress, which may lead to safety issues. In conventional studies, a charging strategy that controls the dc value of each phase current to be the same has been proposed to suppress rotation [8]–[10]. In contrast, this paper proposes an efficiency-enhanced charging method utilizing only two-phases, ensuring zero-torque.

Since the integrated charging method utilizes the stator windings, the periodic fluctuation in phase currents generates core losses. Therefore, reducing core loss is an effective way to enhance the efficiency. In [11], [12], it is shown that core loss decreases as the current ripple diminishes. To reduce the ripple, the conventional strategy adopts three-phase interleaved switching, which is particularly effective when the voltage gain is equal to 1.5 and 3. In contrast, as the industry requires double-voltage boosting, this paper proposes a two-phase interleaved dc charging strategy. This paper theoretically analyzes the phase current ripple regarding voltage gain and verifies the efficacy of the proposed method with experimental results.

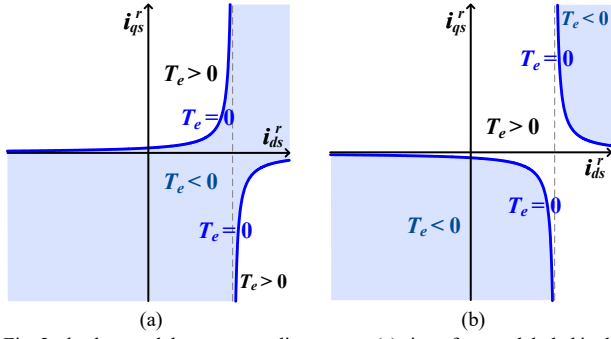


Fig. 2. dq plane and the corresponding torque. (a) sign of torque labeled in dq plane when $\partial\psi/\partial\theta_r$ is positive (b) when negative

II. ANALYSIS OF ZERO-TORQUE

In the charging mode, the sum of each phase current is nonzero since the neutral point of the motor is utilized. As shown in Fig. 1, the charger current is separated into phases, and the zero-sequence current (ZSC) is derived:

$$i_{ns} = \frac{i_{as} + i_{bs} + i_{cs}}{3} = -\frac{i_{charger}}{3} \quad (1)$$

where i_{ns} is the ZSC, i_{abcs} is the a , b , and c phase stator currents, and $i_{charger}$ is the charger current. Unlike in traction mode, zero-sequence components exist and affect the net torque:

$$\begin{aligned} T_e &= \frac{P_m}{\omega_r/pp} = \frac{e_{as}i_{as} + e_{bs}i_{bs} + e_{cs}i_{cs}}{\omega_r/pp} \\ &= 1.5 \frac{e_{ds}^r i_{ds}^r + e_{qs}^r i_{qs}^r + 2e_{ns}^r i_{ns}^r}{\omega_r/pp} \end{aligned} \quad (2)$$

where P_m is the mechanical power of the motor, ω_r is the electrical angular speed, pp is the number of pole pairs, e_{dqns}^r and i_{dqns}^r are the dqn -axis back electromotive force (EMF) and stator current in rotor reference frame, respectively [13], [14]. In addition, the back EMF can be expressed as:

$$\begin{cases} e_{ds}^r = -\omega_r L_q i_{qs}^r \\ e_{qs}^r = \omega_r L_d i_{ds}^r + \omega_r \psi_f \\ e_{ns}^r = \frac{d\psi_n}{dt} \end{cases} \quad (3)$$

where L_d and L_q are the d and q axis inductance of the stator windings, ψ_f is the permanent magnet flux linkage, and ψ_n is the zero-sequence flux linkage. Putting (3) into (2), the following is obtained:

$$T_e = 1.5pp\{\psi_f + (L_d - L_q)i_{ds}^r\}i_{qs}^r + 3pp\frac{\partial\psi_n}{\partial\theta_r}i_{ns}^r \quad (4)$$

where θ_r is the rotor position. Depending on the contribution of each axis current, T_e can be separated into T_{dq} and T_n as:

$$T_{dq} = 1.5pp\{\psi_f + (L_d - L_q)i_{ds}^r\}i_{qs}^r \quad (5)$$

$$T_n = 3pp\frac{\partial\psi_n}{\partial\theta_r}i_{ns}^r. \quad (6)$$

It is noted that torque is also generated by the ZSC and the angular derivative of the zero-sequence flux.

From (4), the zero-torque condition is derived and is presented as:

$$-\frac{T_e}{1.5pp(L_q - L_d)} = \left(i_{ds}^r - \frac{\psi_f}{L_q - L_d}\right)i_{qs}^r - \frac{2i_{ns}^r}{L_q - L_d}\frac{\partial\psi_n}{\partial\theta_r} = 0. \quad (7)$$

When i) i_{ns}^r is determined by $i_{charger}$ and ii) parameters ψ_f , L_d , and L_q are given, the set of zero-torque points form a hyperbola in the dq plane, as illustrated in Fig. 2. To maintain zero-torque, current regulation is required to direct $\vec{i}_{dqs}$ onto the hyperbolic curve. The main principle of the proposal is to regulate the phase current vector along the curve, thus suppressing rotation.

The conventional charging strategy controls the dc value of each phase current to be identical. Meanwhile, $i_{as} = i_{bs} = i_{cs}$ corresponds to $i_{ds} = i_{qs} = 0$. From (5) and (6), the current vector $(i_{ds}^r, i_{qs}^r) = (0, 0)$ eliminates T_{dq} , yet T_n is still present. Although the conventional method generates zero-sequence torque, the resistive torque induced by gear stress, coulomb friction, and the parking brake opposes it. Hence, in practical applications, the conventional method keeps the motor at a standstill. On the contrary, with the proposed strategy, the current vector is directly regulated to a stable zero-torque point with distinct phase current values. As a result, the proposed method explicitly eliminates T_e by taking zero-sequence torque into consideration.

III. PROPOSED TWO-PHASE INTERLEAVED DC CHARGING STRATEGY

A. Two-phase dc charging

This section analyzes the phase current combinations that satisfy both zero-torque and two-phase conduction conditions. Without loss of generality, let the a and b phase currents charge the battery. Then, the following condition (8) is satisfied.

$$\begin{cases} i_{as} + i_{bs} = -i_{charger} \\ i_{cs} = 0 \end{cases} \quad (i_{as}, i_{bs}, i_{cs} \leq 0) \quad (8)$$

The Park transformation is applied to express (8) in the stationary reference dq plane as:

$$\begin{bmatrix} i_{ds}^s \\ i_{qs}^s \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \sqrt{3} & -\sqrt{3} \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -2i_{charger} - 3i_{bs} \\ \sqrt{3}i_{bs} \end{bmatrix}. \quad (9)$$

The set of (i_{ds}^s, i_{qs}^s) from (9) corresponds to a linear function in the dq plane, as presented in (10).

$$\begin{aligned} i_{qs}^s &= -\frac{1}{\sqrt{3}}\left(i_{ds}^s + \frac{2}{3}i_{charger}\right) \\ \left(i_{ds}^s \in \left[-\frac{2i_{charger}}{3}, \frac{i_{charger}}{3}\right], i_{qs}^s \in \left[-\frac{i_{charger}}{\sqrt{3}}, 0\right]\right) \end{aligned} \quad (10)$$

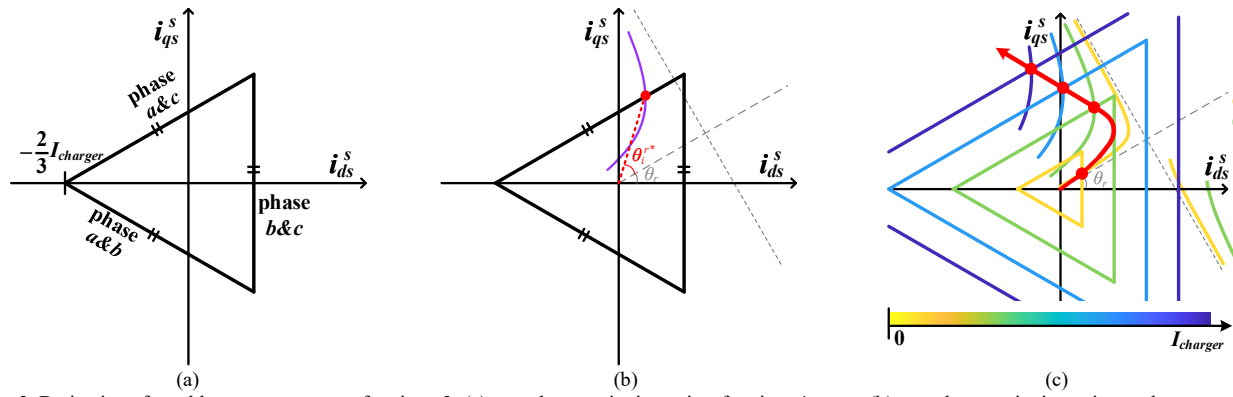


Fig. 3. Derivation of a stable zero-torque curve for given θ_r . (a) two-phase excitation points for given $i_{charger}$ (b) two-phase excitation points and zero-torque points. The intersection is marked in red (c) intersections form a zero-torque curve for given θ_r

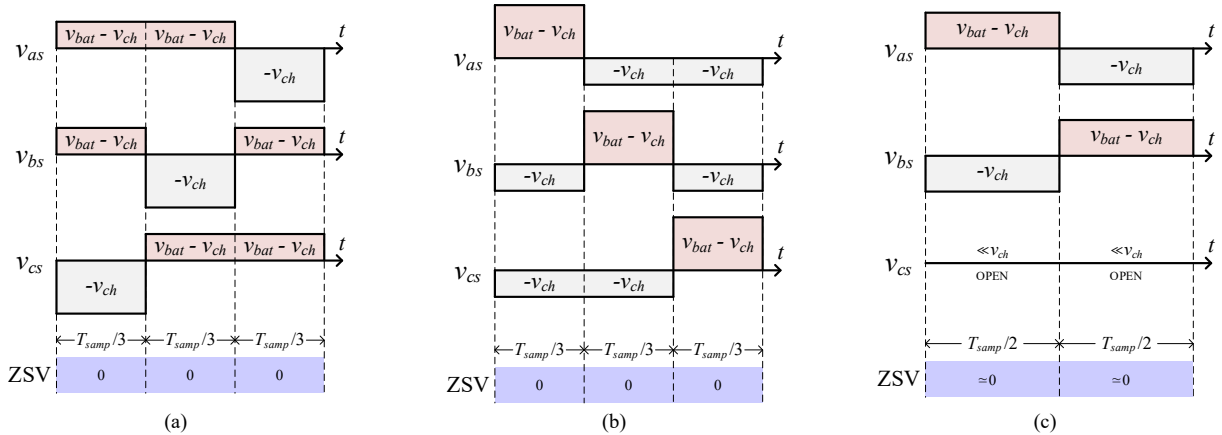


Fig. 4. Ideal phase voltages and the corresponding ZSV in different gains ($G_v = v_{bat}/v_{ch}$). (a) $G_v = 1.5$, three-phase interleaved switching (b) $G_v = 3$, three-phase interleaved switching (c) $G_v = 2$, two-phase interleaved switching

In the same way, distinct linear functions are derived for all combinations of selecting two windings. To sum up, the set of (i_{ds}^s, i_{qs}^s) that satisfy two-phase conduction for a given $i_{charger}$ forms a triangle in the dq plane, as shown in Fig. 3 (a).

Fig. 3 (b) combines Fig. 3 (a) and Fig. 2 into the same plot, highlighting the intersections marked in red; those achieve zero-torque with two-phase excitation, simultaneously. For a given $i_{charger}$ and θ_r , regulating the current to the intersection ensures rotor stability, even though only two phases are utilized.

As presented in Fig. 3 (a), the two-phase excitation triangle expands as $i_{charger}$ increases. In addition, as noted in (7), the gap between the two branches of a hyperbola widens as $i_{charger}$ increases. Consequently, the intersection of the hyperbola and the triangle shifts as $i_{charger}$ changes. With the variation of $i_{charger}$, the set of intersections forms a curve for a given θ_r , as illustrated in Fig. 3 (c). Therefore, the current reference of each winding is determined according to the curve, as the charger current reference varies.

B. Interleaved switching

The integrated charging technique causes repetitive changes in phase currents, resulting in core loss. Meanwhile, studies have indicated that a decrease in current ripple corresponds to a reduction in core loss. Hence, this section analyzes the switching technique for minimizing current ripple, which in turn reduces

core loss. Conventionally, 120° interleaved switching has been employed to reduce the current ripple. On the other hand, this paper proposes a two-phase charging strategy with 180° interleaved switching. For the analysis, the common-mode and the differential-mode components of v_{abcs} and i_{abcs} are studied.

The stator inductance L_{abcs} and the flux linkage λ_{abcs} satisfy the following:

$$L_{abcs} = L_{ls}I + \frac{L_{ds} + L_{qs} - 2L_{ls}}{6} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} + \frac{L_{ds} - L_{qs}}{3} \begin{bmatrix} \cos(2\theta_r) & \cos 2\left(\theta_r - \frac{\pi}{3}\right) & \cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ \cos 2\left(\theta_r - \frac{\pi}{3}\right) & \cos 2\left(\theta_r + \frac{\pi}{3}\right) & \cos(2\theta_r) \\ \cos 2\left(\theta_r + \frac{\pi}{3}\right) & \cos(2\theta_r) & \cos 2\left(\theta_r - \frac{\pi}{3}\right) \end{bmatrix} \quad (11)$$

where L_{ls} is the stator leakage inductance, L_{abcs} is the stator inductance matrix, L_{ns} is the zero-sequence inductance [15]. For fixed θ_r , the stator voltage v_{abcs} is represented as:

$$v_{abcs} \cong \frac{d\lambda_{abcs}}{dt} = \left(L_{abcs} + \frac{L_{ns}}{3} J_3 \right) \frac{di_{abcs}}{dt} \quad (12)$$

where J_3 is a 3-by-3 all-ones matrix. For the analysis, the stator voltage and the phase current are separated into common-mode

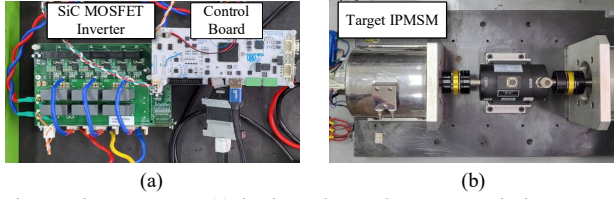


Fig. 5. Laboratory setup. (a) the three-phase voltage-source SiC inverter and the control board (b) the target IPMSM

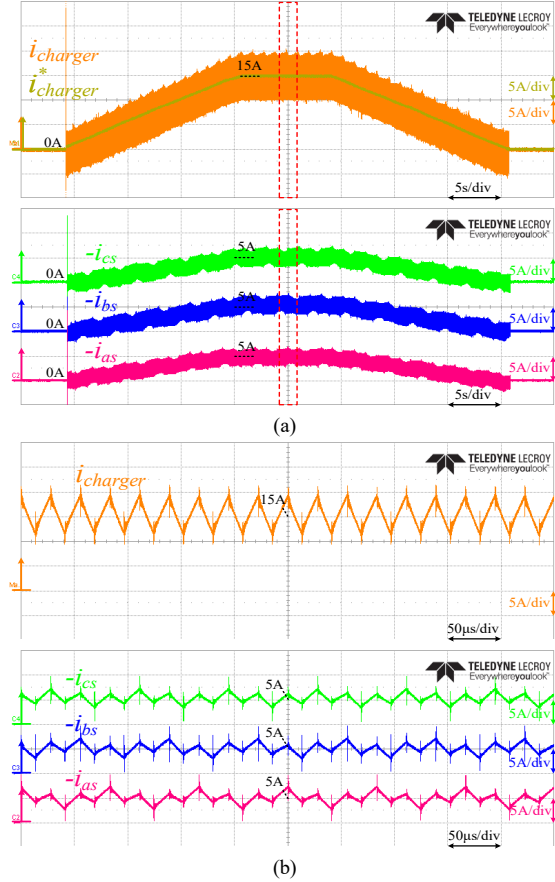


Fig. 6 Experimental results of the conventional method as $i_{charger}^*$ changes when $\theta_{r,init} = -0.09$ rad, $f_{sw} = 12$ kHz, $V_{battery} = 400$ V, and $V_{charger} = 200$ V. (a) $i_{charger}$ and i_{abcs} (b) the steady-state waveforms of $i_{charger}$ and i_{abcs}

and differential-mode components. From (12), the common-mode and the differential-mode inductances are derived as:

$$\begin{aligned} \mathbf{v}_{abcs,CM} &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} v_{CM} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \left(L_{ls} + \frac{L_{ns}}{3} \right) \frac{di_{CM}}{dt} \\ \mathbf{v}_{abcs,DMd} &= \begin{bmatrix} \cos(\theta_r) \\ \cos\left(\theta_r - \frac{2\pi}{3}\right) \\ \cos\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix} v_{DMd} = \begin{bmatrix} \cos(\theta_r) \\ \cos\left(\theta_r - \frac{2\pi}{3}\right) \\ \cos\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix} L_d \frac{di_{DMd}}{dt} \\ \mathbf{v}_{abcs,DMq} &= \begin{bmatrix} \sin(\theta_r) \\ \sin\left(\theta_r - \frac{2\pi}{3}\right) \\ \sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix} v_{DMq} = \begin{bmatrix} \sin(\theta_r) \\ \sin\left(\theta_r - \frac{2\pi}{3}\right) \\ \sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix} L_q \frac{di_{DMq}}{dt}. \end{aligned} \quad (13)$$

TABLE I
EXPERIMENTAL PARAMETERS

Symbol	Value	Symbol	Value
$V_{battery}$	400 V	L_q	4.54 mH
$V_{charger}$	200 V	L_d	7.66 mH
f_{sw}	12 kHz	L_{ls}	0.15 mH
pp	3	ψ_f	0.079 V·s/rad
R_s	0.345 Ω	P_{rated}	3 kW

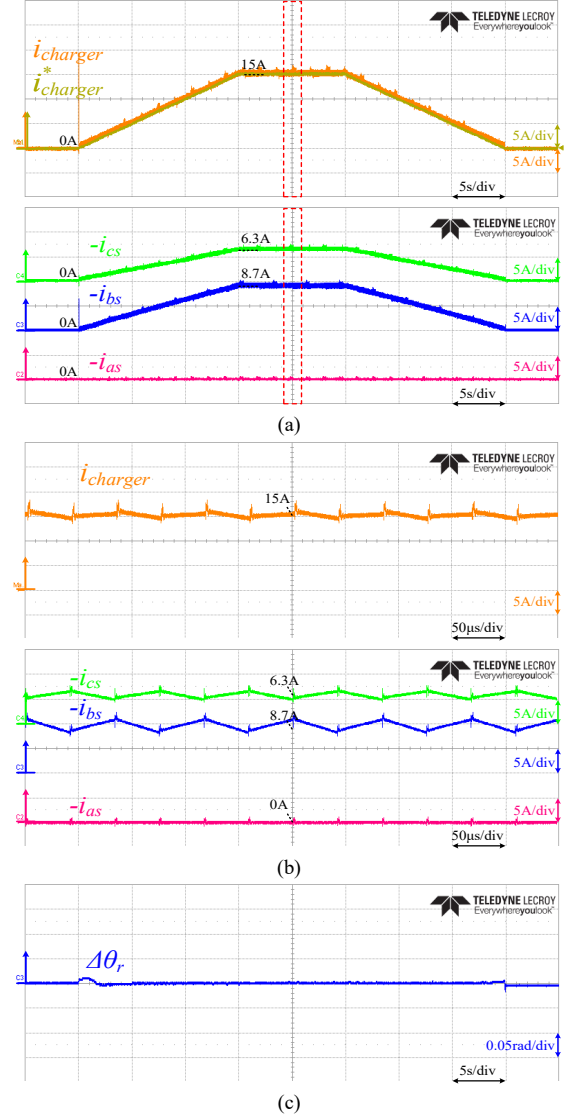


Fig. 7 Experimental results of the proposed method as $i_{charger}^*$ changes when $\theta_{r,init} = -0.09$ rad, $f_{sw} = 12$ kHz, $V_{battery} = 400$ V, and $V_{charger} = 200$ V. (a) $i_{charger}$ and i_{abcs} (b) the steady-state waveforms of $i_{charger}$ and i_{abcs} (c) the rotational movement

As demonstrated, the common-mode inductance is $L_{ls} + L_{ns}/3$, and the differential-mode inductances in d and q axes are L_{ds} and L_{qs} , respectively.

Generally, $L_{qs} > L_{ds} \gg L_{ls} + L_{ns}/3$ is satisfied for interior permanent magnet synchronous motors (IPMSMs), hence the differential-mode inductance is greater than the common-mode

inductance. Since the time derivative of the current is inversely proportional to the inductance for a given applied voltage, the generation of phase current ripple is predominantly attributed to the zero-sequence voltage (v_{CM}). Therefore, diminishing v_{CM} at any instant must be prioritized to reduce the current ripple. In other words, reducing $|v_{as} + v_{bs} + v_{cs}|$ effectively suppresses the ripple.

Fig. 5 illustrates the ideal phase voltages and the corresponding ZSV in different voltage gains ($G_v = v_{bat}/v_{ch}$), neglecting stator resistance. As the integrated charger configuration is identical to three pairs of boost converters, each phase follows the switching pattern of a boost converter. As demonstrated in Fig. 5 (a) and (b), the conventional method with three-phase interleaved switching eliminates ZSV when the voltage gain of the converter equals to 1.5 and 3. In contrast, this paper proposes two-phase interleaved switching, which suppresses ZSV in a double-voltage configuration, as shown in Fig. 5 (c). Since the industry requires a twofold increase in voltage, the proposed method is more effective at reducing current ripple than the conventional one.

IV. EXPERIMENTAL RESULTS

The validity of the proposed method is verified by the experiment. In practice, the goal is to charge 800 V EV battery packs using chargers with a maximum voltage of 410 V. Due to the voltage limit of the laboratory setup, the experiments are conducted in a scaled-down setting with $(v_{bat}, v_{ch}) = (400 \text{ V}, 200 \text{ V})$. The laboratory setup is presented in Fig. 5 and the parameters are listed in Table I. For the inverter, silicon carbide (SiC) MOSFETs (C3M0060065K by Wolfspeed) are used. This chapter demonstrates i_{ch} , i_{abcs} , and $\Delta\theta_r$ as i_{ch}^* changes from 0 to 15 A (i_{ch}^* at rated power).

Fig. 6 (a) and Fig. 7 (a) depict the current waveforms of the full response when using the conventional and the proposed methods, respectively. In addition, Fig. 6 (b) and Fig. 7 (b) are the zoomed-in waveforms in steady-state, at the rated power condition. The switching frequency, f_{sw} , is set to 12 kHz. Fig. 6 indicates that conventional three-phase interleaved switching resulted in noticeable current ripples. In contrast, the current ripples are significantly reduced by the proposed strategy. As studied in section III-B, $|v_{as} + v_{bs} + v_{cs}|$ is suppressed by two-phase interleaved switching in double-voltage setup, thus reducing current ripples. However, conventional switching generates nonzero ZSV, thereby inducing common-mode ripple current. Moreover, Fig. 7 (c) demonstrates that the proposed method effectively impedes rotation during charging. As analyzed in section III-A, the proposed method regulates the current vector along the zero-torque curve, thereby achieving zero-torque and two-phase conduction at the same time.

Fig. 8 compares the charging efficiency of the conventional and the proposed strategies, as the switching frequency alters from 10 kHz to 30 kHz. As presented, the efficiency is enhanced by the proposed method across all tested frequencies. As analyzed in section III-B, the two-phase interleaved switching increases efficiency by reducing the ripple current, which is also presented in Fig. 6 and Fig. 7. Since core loss is attributed to the ripple current, efficiency is enhanced by the proposed strategy. One of the advantages of the proposed method is that it does not

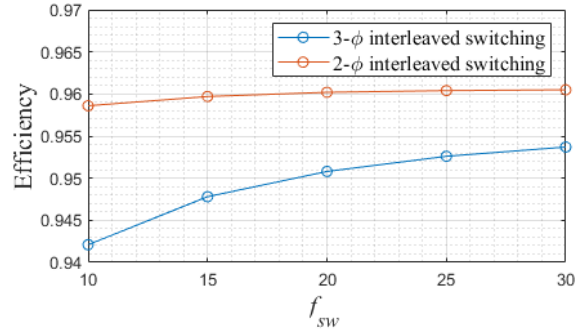


Fig. 8. Charging efficiency of the conventional and the proposed strategy at rated power condition (=3 kW), as the switching frequency varies from 10 kHz to 30 kHz. $P_{ch} = 3 \text{ kW}$, $\theta_r = 0 \text{ rad}$, $V_{bat} = 400 \text{ V}$, and $V_{ch} = 200 \text{ V}$.

require a high switching frequency. Within the frequency range, the efficiency increases as the frequency elevates. In case of the conventional method, the core loss reduction due to an increase in frequency is relatively significant. However, even with 30 kHz switching, the efficiency reaches 95.4%. On the other hand, the proposed method achieved the efficiency of 95.9% even at 10 kHz. This demonstrates that higher efficiency can be obtained by the proposed strategy, even if the system consists of devices unsuitable for high-frequency switching.

V. CONCLUSION

Recently, there has been a noticeable increase in the battery voltage of EVs, marking a twofold increase from the previously standardized models. However, this change has brought a challenge in using conventional fast dc chargers, which require double-voltage dc-dc conversion to charge the battery. To deal with this matter, the integrated charger that utilizes the traction system has been studied and industrialized.

Meanwhile, since the integrated charger utilizes stator current to charge the battery, it is essential to keep the rotor at a standstill to prevent safety issues caused by unintended torque. To avoid rotational movement while charging, the conventional strategy controls the dc value of each phase current to be identical. On the other hand, this paper proposed a method to suppress rotational movement by utilizing only two phases. Hence, this paper analyzes the zero-torque and two-phase excitation conditions. The analyses reported that the current vector must be regulated along the zero-torque curve, which is the set of intersections between zero-torque and two-phase excitation, as presented in Fig. 3 (c). In the integrated charger, diminishing core loss is an effective way to enhance efficiency. The analysis demonstrated that suppressing zero-sequence voltage reduces ripple current and thus the core loss. Therefore, the two-phase interleaved switching proposed in this paper makes the charging protocol more energy efficient.

The experimental results validated the analyses. With the proposed control strategy, current ripples are noticeably reduced compared to the conventional method, and the rotor displacement is also suppressed. Due to the reduced current ripple, the proposed method exhibited higher efficiency than the conventional one, across the switching frequency range from 10 kHz to 30 kHz.

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